Hekai Bu

Wuhan, China
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EDUCATION

Wuhan University Sep 2023 - Jun 2026
Master of Solid Mechanics Weighted Average Score: 85/100
Research Areas: Molecular Dynamics, High Performance Computing, Two Dimensional Materials

Wuhan University Sep 2019 - Jun 2023
Bachelor of Engineering Mechanics Weighted Average Score: 88/100
Awards: Wuhan University Second-Class Scholarship (2021); Wuhan University Outstanding Student Award (2021); Provincial Second Prize, National College Students' Mathematics Competition (2020)

ACADEMIC RESEARCHES

Scalable Modeling Framework for Large-scale vdW Heterostructures Jun 2023 - May 2026
Developer, Researcher

- Developed a hybrid sMLP+ILP framework that achieves near-DFT accuracy with empirical-method efficiency, reducing required training configurations by over **10×**.
- Achieved NEP+ILP approach in LAMMPS (CPU) by updating the NEP_CPU interface, and implemented a highly combined hybrid potential NEP+ILP in GPUMD (GPU).
- Validated mechanical (Young's modulus, Poisson's ratio) and thermal (phonon spectrum, thermal conductivity) predictions for graphene, h-BN and TMDs against experimental benchmarks.

WORK EXPERIENCE

YUSUR Technology Co., Ltd. Wuhan, China
C++ Software Engineer — Intern Mar 2023 - Jun 2023

- Implemented QDMA in QEMU for host-to-card transfers: descriptor packaging, DMA execution, status-queue management, and packet validation via ping.
- Built a Docker environment and debugged the open-nic driver on a hardware simulator.
- Collaborated on coding standards and managed version control using GitLab.
- Compiled application documentation and drafted software copyright and patent registrations.

PROJECTS

GPUMD and NEP: Performance Optimization & Feature Expansion Jul 2023 - Present
Key Developer

- Profiled GPU kernels with NVIDIA Nsight Compute and refactored bottleneck functions by optimizing algorithms to achieve **15%** speedup (GPUMD: PR 762, PR 806).
- Integrated 3 new hybrid potentials for vdW materials: SW+ILP, Tersoff+ILP, NEP+ILP (GPUMD: PR 761, PR 1062, PR 951).
- Maintained NEP_CPU repository and enhanced the NEP–LAMMPS interface to support hybrid potentials in LAMMPS (NEP_CPU: PR 34).
- Developed a ghost atom computation module for the GPUMD Deep Potential interface, boosting throughput and reducing GPU memory usage (GPUMD: PR 827).
- Updated the official GPUMD user manual and maintained internal developer documentation (GPUMD: PR 1068, PR 791).

PUBLICATIONS

[1] **H. Bu**#, W. Jiang#, P Ying, T. Liang, Z. Fan, W. Ouyang, Modular hybrid machine learning and physics-based potentials for scalable modeling of Van der Waals heterostructures, J. Mech. Phys. Solids 2026, 210, 106540. [Co-first author]
[2] W. Jiang#, **H. Bu**#, T. Liang#, P. Ying, Z. Fan, J. Xu, W. Ouyang, Accurate modeling of interfacial thermal transport in van der Waals heterostructures via hybrid machine learning and registry-dependent potentials, J. Chem. Theory Comput. 2026, 22(9):4699-4715. [Co-first author]
[3] K. Xu, **H. Bu**, S. Pan, E. Lindgren, Y. Wu, Y. Wang, J. Liu, K. Song, B. Xu, Y. Li, T. Hainer, L. Svensson, J. Wiktor, R. Zhao, H. Huang, et. al., P. Guan, P. Erhart, J. Sun, W. Ouyang, Y. Su, Z. Fan, GPUMD 4.0: a high-performance molecular dynamics package for versatile materials simulations with machine-learned potentials, MGE Advances. 2025, 3 (3), e70028.
[4] W. Jiang, **H. Bu**, B. Wu, W. Ouyang, Chirality-dependent strain modulation of thermal conductivity in double-walled

molybdenum disulfide nanotubes, Int. Commun. Heat Mass Transf. 2026, 170, 109935.
[5] W. Jiang, T. Liang, **H. Bu**, J. Xu, W. Ouyang, Moiré-driven interfacial thermal transport in twisted transition metal dichalcogenides, ACS Nano 2025, 19 (17), 16287–16296.
[6] T. Liang#, W. Jiang#, K. Xu, **H. Bu**, Z. Fan, W. Ouyang, J. Xu, PYSED: A tool for extracting kinetic-energy-weighted phonon dispersion and lifetime from molecular dynamics simulations, J. Appl. Phys. 2025, 138 (7), 075101.

SOFTWARE COPYRIGHTS

- Z. Liu, W. Ouyang, **H. Bu**, High-Performance Software for Object Matching and Tracking Based on Temporal Image Feature Analysis, 2023SR0562225, National Copyright Administration, May 2023.

SKILLS

- **Programming Language:** C & C++, CUDA C++, Python, Matlab, Fortran
- **Development Tools:**: Git, GDB, CUDA-GDB, Make, Nsight Compute, Vim, SLURM
- **Academic Software:** GPUMD, LAMMPS, VASP, Ovito, Adobe Illustrator
- **Language:** : Chinese (native), English (CET-6)